Problem solving and testing using java

WEEK 1 TASKS

20/7/2026-25/7/2026

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Task 1: Access and print the element at a given index in an array.

Task 2: Search for a given element in a sorted array using Binary Search.

Task 3: Find the maximum element in an array of n integers.

Task 4: Find the Kth smallest element in an array.

Task 5: Print all possible pairs of elements from an array.

Task 6: Sum of even or odd digits.

Task 7: Nth Fibonacci.

Task 8: Check whether a number is a Palindrome.

Task 9: Sum of the last digits of two given numbers.

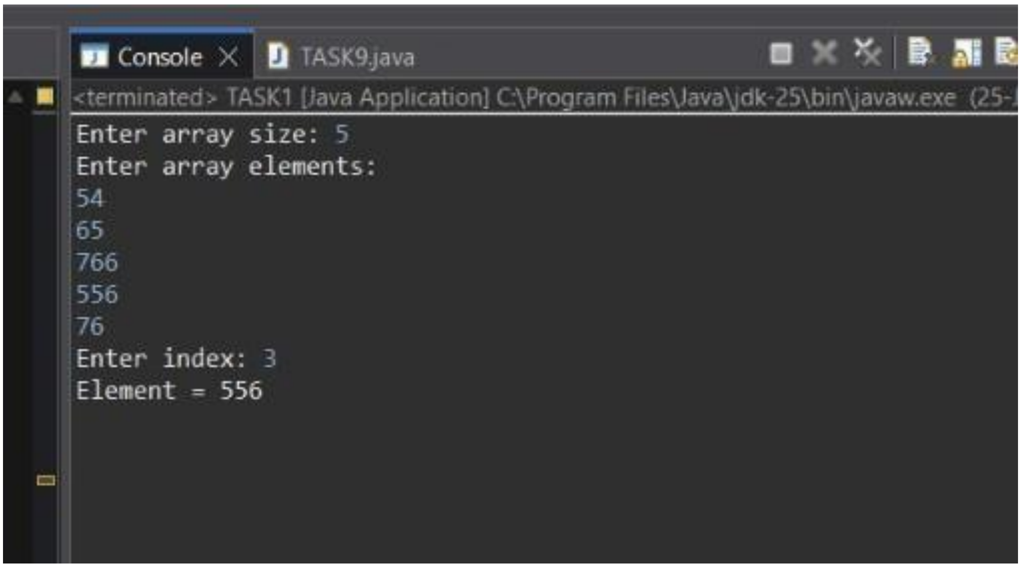
TASK 1

```
package lambda_function; import

java.util.Scanner;
```

```
@FunctionalInterface      interface
ArrayElement {    void display(int[]
arr, int index);
}

public class TASK1 {     public static void
main(String[] args) {

       Scanner sc = new Scanner(System.in);

       System.out.print("Enter    array    size:    ");
int n = sc.nextInt();

       int arr[] = new int[n];

       System.out.println("Enter array elements:");
       for  (int  i  =  0;  i  <  n;  i++)  {
arr[i] = sc.nextInt();
       }

       System.out.print("Enter     index:     ");
int index = sc.nextInt();

       //       Lambda       Expression
ArrayElement  obj  =  (a,  i)  ->  {
if (i >= 0 && i < a.length)
              System.out.println("Element    =    "    +    a[i]);
else
```

```
        System.out.println("Invalid Index, that index is not there");
    };

    obj.display(arr, index);

//      sc.close();
  }
}
```

OUTPUT:



TASK 2 package

lambda_function; import

java.util.Scanner;

@FunctionalInterface interface

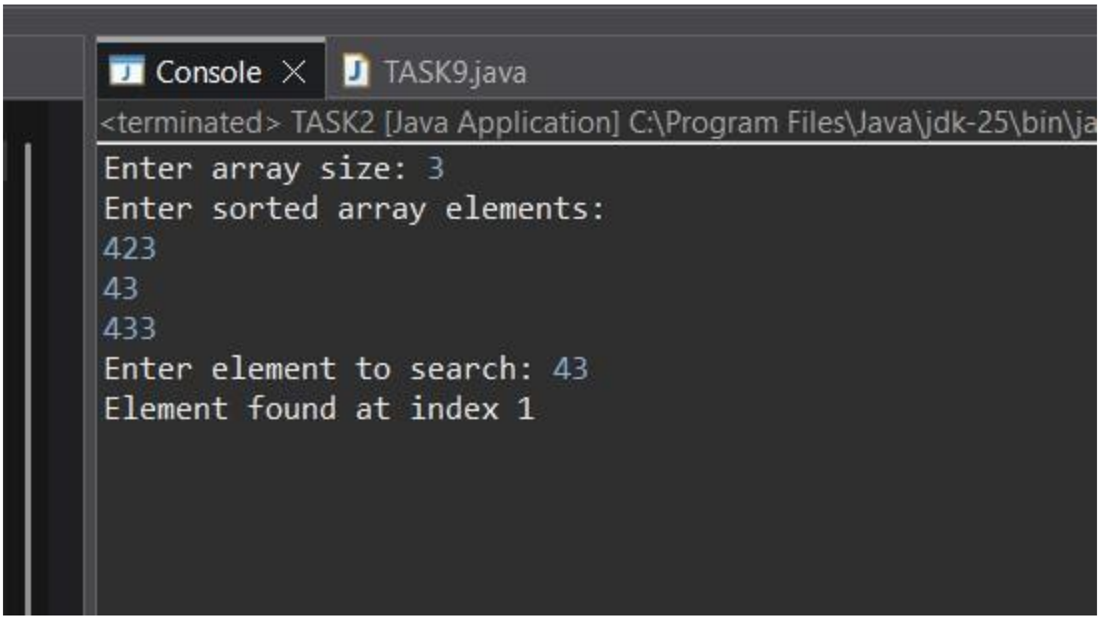
BinarySearch { void

search(int[] arr, int key);

```
}
public class TASK2 {      public static void
main(String[] args) {
        Scanner sc = new Scanner(System.in);
System.out.print("Enter array size: ");        int
n = sc.nextInt();        int arr[] = new int[n];
        System.out.println("Enter sorted array elements:");
        for  (int  i  =  0;  i  <  n;  i++)  {
arr[i] = sc.nextInt();
        }
        System.out.print("Enter   element   to   search:   ");
int key = sc.nextInt();
        // Lambda Expression
BinarySearch bs = (a, k) -> {
int low = 0;            int high =
a.length - 1;            boolean
found = false;            while (low
<= high) {                int mid = (low
+ high) / 2;


                if (a[mid] == k) {
                    System.out.println("Element found at index " + mid);
found = true;                    break;
                }  else  if  (a[mid]  <  k)  {
low = mid + 1;
                }        else        {
high = mid - 1;
                }
```

```
        }
        if (!found) {
            System.out.println("Element not found");
        }
    };
    bs.search(arr,                    key);
sc.close();
  }
}
```

OUTPUT:

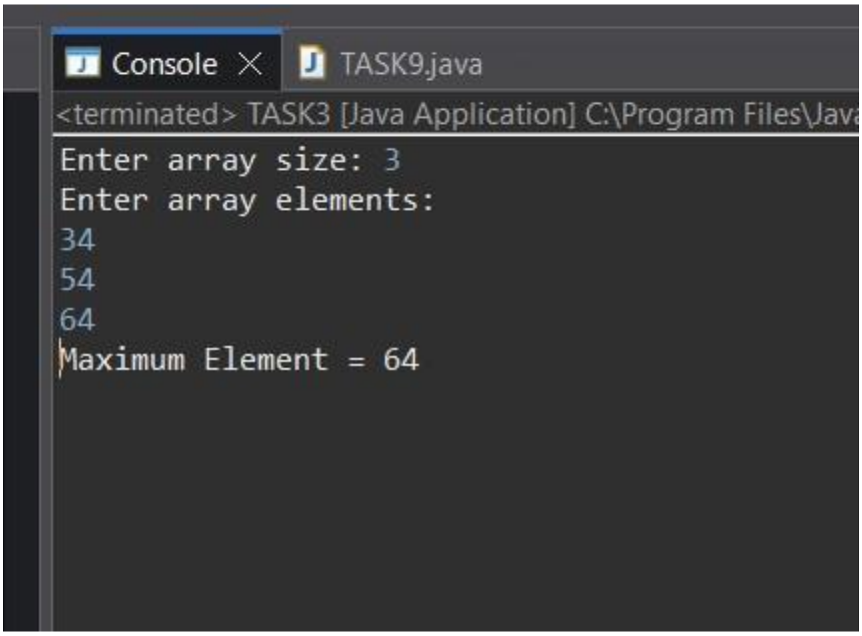



```
TASK        3        package
lambda_function;     import
java.util.Scanner;
@FunctionalInterface
interface Maximum {
  void findMax(int[] arr);
}
```

```
public class TASK3 {      public static void
main(String[] args) {
        Scanner sc = new Scanner(System.in);
System.out.print("Enter array size: ");        int
n = sc.nextInt();        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for  (int  i  =  0;  i  <  n;  i++)  {
arr[i] = sc.nextInt();
        }
        //          Lambda          Expression
Maximum maxElement = (a) -> {
            int max = a[0];            for (int i
= 1; i < a.length; i++) {                if
(a[i] > max) {                    max = a[i];
                }
            }

            System.out.println("Maximum Element = " + max);        };
        maxElement.findMax(arr);
sc.close();
    }
}
```

OUTPUT:

}



TASK 4 package lambda_function;

import java.util.Arrays; import java.util.Scanner;

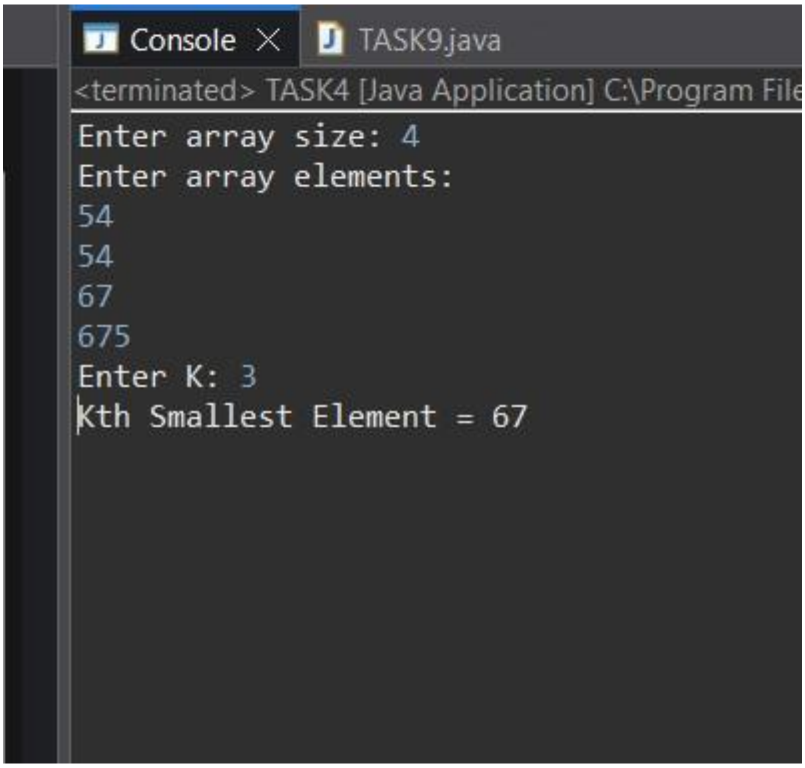
@FunctionalInterface interface KthSmallestElement { void findKthSmallest(int[] arr, int k);

}

```
public class TASK4 {      public static void
main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter   array   size:   ");
int n = sc.nextInt();

        int arr[] = new int[n];

        System.out.println("Enter array elements:");
        for  (int  i  =  0;  i  <  n;  i++)  {
arr[i] = sc.nextInt();
        }

        System.out.print("Enter    K:    ");
int k = sc.nextInt();

        // Lambda Expression
        KthSmallestElement obj = (a, key) -> {           Arrays.sort(a);

            if (key >= 1 && key <= a.length) {
                System.out.println("Kth Smallest Element = " + a[key - 1]);
            } else {
                System.out.println("Invalid K");
            }
        };
```

```
        obj.findKthSmallest(arr, k);

        sc.close();
    }
}
```

OUTPUT :



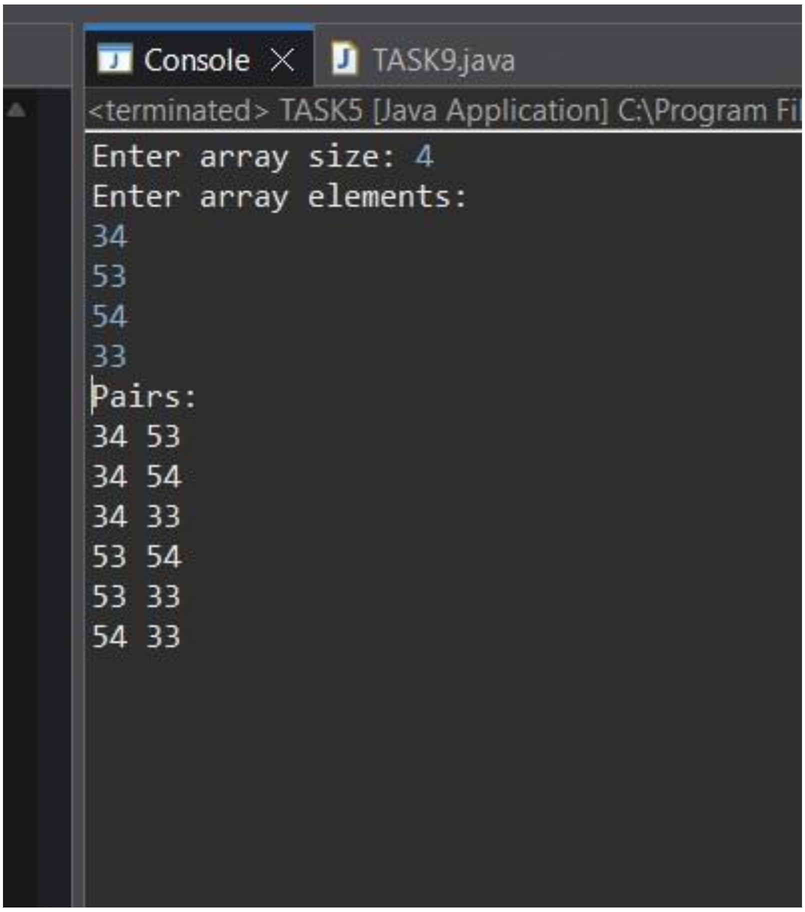
TASK 5 package lambda_function; import java.util.Scanner;

```
@FunctionalInterface
interface ArrayPair {      void
printPairs(int[] arr);
}
public class TASK5 {
```

```
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter    array    size:    ");
int n = sc.nextInt();

        int arr[] = new int[n];

        System.out.println("Enter array elements:");
        for  (int  i  =  0;  i  <  n;  i++)  {
arr[i] = sc.nextInt();
        }

        //          Lambda          Expression
ArrayPair pair = (a) -> {
            System.out.println("Pairs:");

            for (int i = 0; i < a.length; i++) {
                for (int j = i + 1; j < a.length; j++) {
                    System.out.println(a[i] + " " + a[j]);
                }
            }
        };

        pair.printPairs(arr);

        sc.close();
```

```
    }
}
```

OUTPUT :


Console
TASK9.java
<terminated> TASK5 [Java Application] C:\Program Fi
Enter array size: 4
Enter array elements:
34
53
54
33
Pairs:
34 53
34 54
34 33
53 54
53 33
54 33

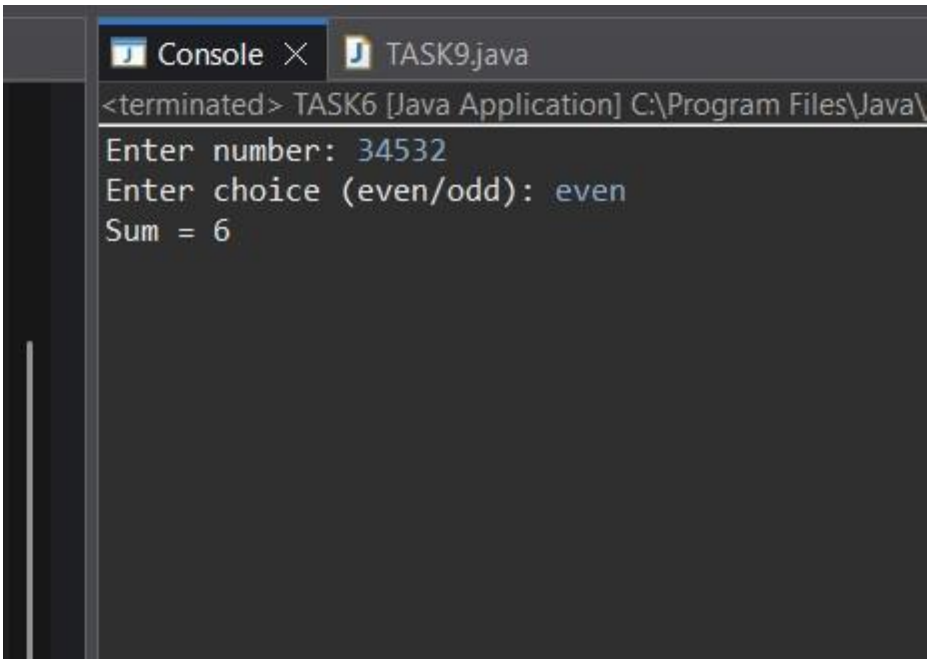
package lambda_function;

TASK 6

```
import java.util.Scanner;
@FunctionalInterface interface DigitSum {
void findSum(int num, String choice);
}
public class TASK6 {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
System.out.print("Enter number: ");        int
num = sc.nextInt();
      System.out.print("Enter choice (even/odd): ");
      String choice = sc.next();

      //        Lambda        Expression
DigitSum ds = (number, ch) -> {
int sum = 0;         while (number > 0)
{            int digit = number % 10;

            if (ch.equalsIgnoreCase("even") && digit % 2 == 0) {
sum += digit;
            }

            if (ch.equalsIgnoreCase("odd") && digit % 2 != 0) {
sum += digit;
            }

            number /= 10;
```

```
        }

        System.out.println("Sum = " + sum);
      };

      ds.findSum(num, choice);

      sc.close();
   }
}
```

OUTPUT

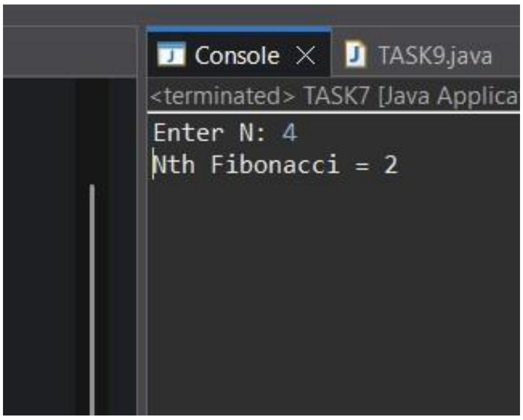


TASK 7

```
package lambda_function;

import java.util.Scanner;


@FunctionalInterface interface
FibonacciNumber {     void
findFibonacci(int n);
}


public class TASK7 {


   public static void main(String[] args) {


       Scanner sc = new Scanner(System.in);


       System.out.print("Enter N: ");
int n = sc.nextInt();


       // Lambda Expression
       FibonacciNumber fib = (num) -> {


          int a = 0, b = 1;


          if (num == 1) {
             System.out.println("Nth Fibonacci = " + a);
          }
          else if (num == 2) {
             System.out.println("Nth Fibonacci = " + b);
          }
```

```
else {
int c = 0;

for (int i = 3; i <= num; i++) {
c = a + b;          a = b;
b = c;
}

System.out.println("Nth Fibonacci = " + c);
}
};
fib.findFibonacci(n);
sc.close();
}
}
```

OUTPUT:



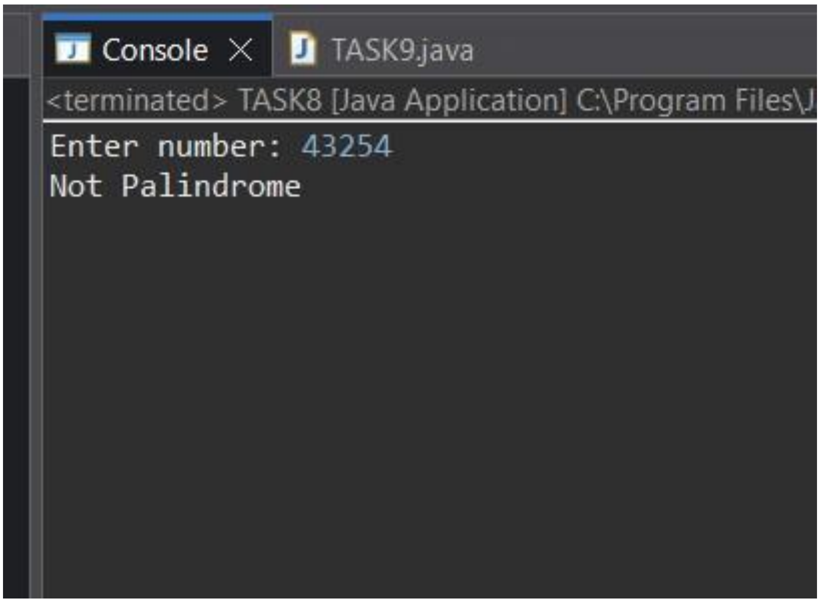
TASK 8

```
import java.util.Scanner;
```

```
package lambda_function;


@FunctionalInterface interface
Palindrome {     void
checkPalindrome(int num);
}

public class TASK8 {

   public static void main(String[] args) {

       Scanner sc = new Scanner(System.in);

       System.out.print("Enter number: ");
int num = sc.nextInt();

       // Lambda Expression
       Palindrome p = (number) -> {

           int original = number;
int reverse = 0;

           while (number > 0) {                int
digit = number % 10;                reverse
= reverse * 10 + digit;               number
/= 10;

           }

           if (original == reverse)
```

```
            System.out.println("Palindrome");
else
            System.out.println("Not Palindrome");
      };


      p.checkPalindrome(num);


      sc.close();
   }
}
```

OUTPUT



```
TASK 9 package
lambda_function;

import java.util.Scanner;

@FunctionalInterface interface
LastDigitSum {     void
sumLastDigits(int a, int b);
}

public class TASK9 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
int a = sc.nextInt();

        System.out.print("Enter second number: ");
int b = sc.nextInt();

        // Lambda Expression
        LastDigitSum sum = (x, y) -> {

            int last1 = x % 10;
int last2 = y % 10;
```

```
System.out.println("Sum of
last digits = " + (last1 +
last2));
        };

        sum.sumLastDigits(a, b);

        sc.close();
    }
}
```

OUTPUT